SOFTWARE QUALITY ENGINEERING

Lab 7: Pairwise Test Design

Book Search Form — 2-Way Covering Array

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Project: LMS Book Search Form
Tool: allpairspy 2.5.1

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1. Objective

Pairwise testing selects a smaller set of test cases so every possible pair of values across every two parameters appears at least once. It reduces the number of test cases compared with testing every possible full combination.

2. Parameters and Values

Parameter	Value 1	Value 2	Value 3
P1 — Query Type	Title	Author	ISBN
P2 — Filter	None	Available	Genre
P3 — Sort Order	Relevance	A-Z	Date
P4 — Page Size	10	20	50

3. Generated Pairwise Test Cases

Test Case	P1: Query	P2: Filter	P3: Sort	P4: Size
PW1	Title	None	Relevance	10
PW2	Author	Available	A-Z	10
PW3	ISBN	Genre	Date	10
PW4	ISBN	Available	Relevance	20
PW5	Author	None	Date	20
PW6	Title	Genre	A-Z	20
PW7	Title	Available	Date	50
PW8	Author	Genre	Relevance	50

PW9	ISBN	None	A-Z	50

Generated Output

- **Generated test cases:** 9
- **Required pairs:** 54
- **Missing pairs:** 0
- **Full combinations:** 81
- **Reduction factor:** 9×

The generated test suite covers all required pairs across the four parameters without requiring exhaustive testing.

4. P1 and P4 Verification

The following table confirms that all nine combinations of query type and page size appear in the generated test cases.

P1 — Query Type	P4 = 10	P4 = 20	P4 = 50
Title	PW1	PW6	PW7
Author	PW2	PW5	PW8
ISBN	PW3	PW4	PW9

Each query type appears with all three page-size values.

5. Removal Experiment

The final test case, PW9 (`ISBN, None, A-Z, 50`), was temporarily removed to evaluate the effect on pairwise coverage.

The script then reported eight test cases and six missing pairs.

Two of the missing pairs were:

Parameter Pair	Missing Value Pair	Reason
P1 and P2	ISBN + None	In the provided generated array, this pair appears only in PW9.
P1 and P4	ISBN + 50	In the provided generated array, this pair appears only in PW9.

After removing the temporary `cases.pop()` line, the script was run again and returned nine test cases, 54 required pairs, and zero missing pairs.

This experiment demonstrates that removing even one test case can reduce pairwise coverage.

6. Reduction Calculation

The total number of exhaustive combinations is calculated as follows:

Full combinations

$3 \times 3 \times 3 \times 3 = 81$

Pairwise test cases

9

Reduction factor

$81 \div 9 = 9$

Number of test cases saved

$81 - 9 = 72$

Percentage reduction

$(72 \div 81) \times 100 \approx 88.9\%$

Therefore, the pairwise test suite uses one-ninth as many test cases as exhaustive testing, reducing the number of test cases by approximately 88.9% while maintaining the required pairwise coverage.

7. Screenshot to Insert

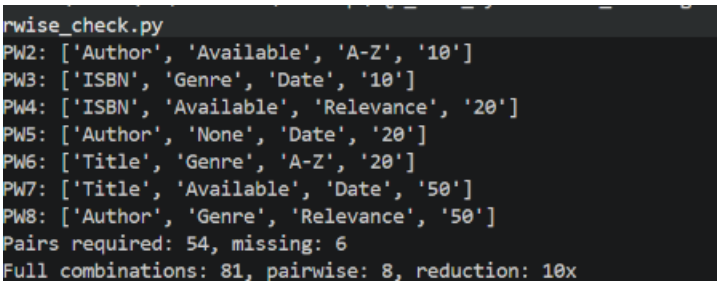
Insert a screenshot of the actual terminal output produced by running `pairwise_check.py`.

The screenshot should show:

- The generated pairwise test cases.
- Total generated test cases: 9.
- Required pairs: 54.
- Missing pairs: 0.
- Full combinations: 81.
- Reduction factor: 9×.

Suggested screenshot filename: `07_Pairwise_Test_Results.png`

Suggested location: `ScreenShots/Pairwise output/`



Conclusion

Pairwise testing was applied to the LMS Book Search Form using four parameters, each containing three possible values. Exhaustive testing would require 81 combinations, whereas the generated pairwise suite requires only nine test cases.

The generated suite covers all 54 required value pairs, with zero missing pairs. The removal experiment demonstrates the importance of individual test cases in maintaining complete pairwise coverage.

The results show that pairwise testing can significantly reduce the number of test cases while providing systematic coverage of interactions between parameter values.